1. Event Tracking & Instrumentation

1. Which events should we implement across **GA4, Meta Pixel, and Conversion API** to track the full ecommerce funnel?

2. How should we design a **standardized event naming schema** that works across GA4, Firebase, Shopify, and MMPs?

3. What parameters should be attached to the **purchase event** so that it supports attribution, revenue tracking, and cohort analysis?

4. Which **GA4 recommended ecommerce events** should we implement for a Shopify store?

5. How should we structure **custom events vs standard events** across different platforms to maintain compatibility with ad platforms?

2. API Data Extraction Strategy

6. Which **Meta Ads API endpoints** should we pull to build a campaign performance dashboard?

7. What API fields should be extracted from **GA4 Data API** to replicate the acquisition reports?

8. How should we structure **Shopify Admin API data pulls** to capture orders, products, customers, and refunds?

9. Which endpoints from **AppsFlyer / Branch / Singular** provide install attribution and post-install events?

10. How should the assistant determine which **API parameters are necessary** for building marketing attribution dashboards?

3. Data Warehouse Schema Design

11. What is the recommended **data warehouse schema** for an ecommerce analytics stack integrating Shopify, GA4, and ad platforms?

12. How should we design tables for **users, sessions, events, and orders** so they can be joined across platforms?

13. What is the best way to structure a **campaign performance table** combining Meta Ads, Google Ads, and TikTok Ads?

14. How should we model **product-level purchase data** to support SKU-level reporting?

15. How should the warehouse schema support **multi-touch attribution analysis**?

4. Attribution & Conversion Mapping

16. Why do **Shopify purchase numbers differ from GA4 purchase events**?

17. Why does **Meta Ads report more conversions than GA4**?

18. How should we reconcile **MMP attribution vs GA4 attribution**?

19. What attribution window differences exist between **Meta Ads, Google Ads, and GA4**?

20. How should we design a **unified attribution model** across ad platforms and analytics tools?

5. Tracking Infrastructure (GTM / SDK / Pixels)

21. How should we structure a **Google Tag Manager container** for scalable ecommerce tracking?

22. How should **Meta Pixel and Conversion API** be implemented together for deduplication?

23. What parameters are required to ensure **event deduplication between Pixel and CAPI**?

24. How should **Firebase SDK events be mapped to GA4 events**?

25. How should we debug cases where **tracking events are firing but not appearing in analytics platforms**?

6. Data Pipeline & ETL Troubleshooting

26. Why did a **Shopify webhook ingestion pipeline fail**?

27. How should we design retry logic for **API rate limits or ingestion failures**?

28. How should we structure **incremental data pulls** for APIs like Meta Ads or Shopify?

29. What strategies can prevent **duplicate event ingestion** in the warehouse?

30. How should we monitor **pipeline failures and missing data intervals**?

7. SQL Query & Data Modeling Support

31. What SQL query should be used to calculate **daily revenue from Shopify orders**?

32. How should we write SQL to compute **ROAS using ad spend and revenue tables**?

33. How should we calculate **customer lifetime value (LTV)** using warehouse data?

34. How should we design SQL queries to calculate **conversion rate across sessions and purchases**?

35. How should we join **ad campaign data with ecommerce purchases** to produce performance reports?

8. Dashboard Architecture (Superset)

36. What metrics should be included in a **growth performance dashboard for ecommerce**?

37. How should a **marketing funnel dashboard** be structured using session, add-to-cart, and purchase data?

38. What is the correct way to calculate **conversion rate across multiple platforms**?

39. How should we design a **campaign performance dashboard** that works across multiple ad networks?

40. How should we structure a **product performance dashboard** using order-level data?

9. Data Freshness & Latency

41. Why is **Shopify order data appearing faster than GA4 purchase data**?

42. What are the typical **data delay windows** for GA4, Meta Ads, and Shopify APIs?

43. How should we design the pipeline to ensure **near-real-time reporting dashboards**?

44. How can we detect **data freshness issues automatically**?

45. What alerts should be triggered when **data ingestion stops or lags behind expected intervals**?

10. Marketing Performance Diagnostics

46. Why did **conversion rate suddenly drop across campaigns**?

47. Why is **ROAS decreasing despite stable traffic volume**?

48. How can we detect **campaign performance anomalies automatically**?

49. Which metrics should be monitored daily to detect **tracking or data issues early**?

50. How should the assistant diagnose **funnel drop-offs between session → add to cart → purchase**?

Example:

Purchase mismatch troubleshooting

Shopify Purchases ≠ GA4 Purchases

Step 1
Check GA4 purchase event implementation

Step 2
Check GTM trigger firing

Step 3
Check consent mode filtering

Step 4
Check attribution windows